COMPUTER VISION
ASSIGNMENT 1

Apply, Compare and Justify Image Acquisition, Formation and Processing Techniques

Vision-Based Mango Grading Line

GreenHarvest Mango Processing Unit | Total: 50 marks

Submission detail	
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Course / section	ITA0523

TASK 1 | 4 MARKS

Overview and features of computer vision: compare grading approaches

Goal: compare manual grading, image-processing assistance and automated machine vision; define a reproducible evaluation rather than inventing human test results.

1.1 Comparison of the three approaches

Criterion	A. Manual grading	B. Image-processing aid	C. Automated computer vision
Input / operation	Worker views moving mango under variable shed light and judges by eye.	Camera image is enhanced and shown to a worker; worker still decides.	Controlled capture + algorithms compute size, ripeness, defects and grade.
Decision consistency	Depends on fatigue, training, viewing angle and ambient light.	May improve visibility but visual judgement remains subjective.	Repeatable for fixed calibration/model; can drift if optics or lighting change.
Speed at 8 fruit/s	One person cannot reliably inspect one fruit every 0.125 s. Parallel staff are required.	Display and operator reaction are bottlenecks; likely slower than direct automated sorting.	Can process asynchronously if capture and decision queues sustain ≥8 fruit/s; measure on target PC.
Non-contact / measurement	Non-contact, but length estimates by eye are coarse.	Non-contact; display zoom does not add detail absent from sensor sampling.	Non-contact; calibrated dimensions may meet ±2 mm if perspective/pose are controlled.
Traceability	Usually handwritten/batch-level records.	Screenshots possible, but records are manual.	Store fruit ID, image ROI, calibration version, features, prediction, confidence and timestamp.
New defect type	Can adapt after training, but agreement must be monitored.	Human can notice unfamiliar signs, depending on attention.	Rule changes or retraining needed; unknown defects can pass without a novelty/fail-safe check.
Costs	Low initial capital; recurring labour and fatigue costs.	Camera, display and software plus labour.	Highest initial integration/calibration cost; lower per-fruit decision cost after deployment.
Best use	Small volumes, visual exceptions, fallback and adjudication.	Training/inspection aid during pilot and human-in-the-loop review.	High-volume, traceable grading after validation and line synchronization.

Recommendation: use approach C for the main grading decision, with approach B as the commissioning and audit interface, and retain human review for uncertain fruit and unknown defect types. A production rule should fail safe: if the fruit mask is incomplete, a confidence threshold is not met, calibration is invalid or an image is blurred, divert the fruit for reinspection rather than automatically accepting it.

1.2 Experimental protocol for 30 mangoes and three evaluators

1. Collect at least 30 one-fruit images, including green/turning/ripe peel, red blush, clean fruit, anthracnose, sap burn, bruises, glossy highlights, stem ends, shadows and partial occlusion. Save the image ID and a ground-truth record agreed by an adjudicator. Thirty images are suitable for a first demonstration, not for demonstrating a 2% miss limit.
2. Manual approach A: three evaluators grade each fruit independently in two rounds separated by a washout interval. Randomize image order and do not show other evaluators' decisions. Record ripeness, defect count, maximum defect diameter, accept/reject and confidence.
3. Image-assisted approach B: show the same evaluators the same images after fixed preprocessing (resize, brightness-normalized view and zoom). Do not permit automatic grade hints. Repeat the same two-round design so the effect of display enhancement can be separated from worker variation.
4. Automated approach C: freeze code, thresholds and calibration before the holdout run. Record per-image processing time, result, confidence and reason codes. Do not tune against holdout labels.
5. Report inter-rater Cohen's κ for each evaluator pair within a round; report intra-rater κ by comparing each evaluator's Round 1 and Round 2 decisions. For three raters, Fleiss' κ can summarize multi-rater agreement. Use separate κ calculations for nominal ripeness and binary defective/not-defective labels.

Cohen kappa: $\kappa = (p_{observed} - p_{expected}) / (1 - p_{expected})$
Throughput: fruits/s = number of completed decisions / timed interval in seconds
Latency: decision timestamp − image-trigger timestamp
Error rate: (false accepts + false rejects) / total adjudicated fruit

The companion script creates `manual_ratings_template.csv` for the 30 synthetic fruit identities. Fill it with human judgements and save it as `manual_ratings.csv` in the script folder; rerunning the script calculates pairwise inter-rater and within-rater Cohen's κ. No human agreement values are reported in this document because no evaluators actually rated the images.

1.3 Processing concepts used in the workflow

Term	Input → output	Role in this workflow
Image processing	Image array → modified image array.	Denoise, flat-field correct, enhance local contrast, resize for display, threshold candidate defects.
Computer vision	Images → measurements, attributes or scene decisions.	Estimate fruit contour, length/width, colour class, spot count and grade.
Machine vision	Integrated sensing + optics + lighting + triggers + vision software + actuator.	The production system includes encoder, camera trigger, PC, I/O and timed air-jet control.
Computer graphics	Scene/model description → rendered image.	Useful for synthetic images, operator UI and design simulation; it is not the measurement sensor itself.

1.4 Manual capacity and unknown defect types

A 10-hour shift has 36,000 seconds. At 8 fruit/s, the line presents 288,000 fruits per shift. A single grader who takes 1 second per fruit handles at most 36,000 fruits per shift, so eight ideal parallel graders are needed just to match arrivals; at 2 seconds per fruit the ideal requirement becomes sixteen. This ignores

fatigue, breaks, variation, record keeping and rework. The 0.125-second average arrival interval is not a realistic per-fruit human decision time.

An unrecognised defect can be false-accepted by a fixed dark-pixel rule, especially if its colour/texture differs from known examples. Corrective measures are: (i) route low-confidence/novel features to human review; (ii) collect confirmed examples and version the rule/model; (iii) run periodic blind audits and challenge-set tests; (iv) log images and feature reasons; and (v) never claim the 2% target based only on familiar defect classes.

1.5 Vision-language model (VLM) test on 10 images

Use the same ten preselected images spanning clean, black-spotted, bruise, sap burn, red blush, glare and shadow. Freeze the model/version, prompt, image size and sampling settings. A suitable prompt is:

> Inspect this single mango image for export grading. Return JSON only with:
> {"ripeness":"green|turning|ripe|uncertain",
> "visible_defects":["anthracnose","sap burn","bruise"],
> "spot_count": integer or null, "evidence": "one sentence",
> "confidence": 0.0 to 1.0}.
> Do not infer internal bruises from a normal RGB image. Use uncertain when evidence is weak.

Comparison dimension	Manual / image-aided	Rule-based CV	VLM (e.g., Florence-2/LLaVA)
Repeatability	Measure κ across people/rounds.	Deterministic if code, input and parameters are fixed.	Repeat ten images at least three times; responses may vary with prompt, decoding and model version.
Explainability	Human can point to visible feature.	Strongest when explanations expose measurements and thresholds.	Natural-language rationale is useful but not guaranteed to be a faithful causal explanation.
Latency	Human response dominates.	Typically suitable for CPU if features/ROIs are small; benchmark exact PC.	Model weights, input size and CPU-only inference can be too slow; measure p50 and p95.
Defect reliability	Can notice unusual cases but can tire.	Weak on unknown defects unless designed to reject uncertain cases.	May describe plausible defects that are not present; must be checked against adjudicated labels.
Export suitability	Good for review / adjudication.	Good main decision path after validation.	Research/triage aid only until a representative test demonstrates required false-accept performance.

No VLM accuracy or latency is claimed: the runtime/model and the ten test images were not provided. Record exact per-image predictions and evaluate defect recall, false acceptance, class-wise F1, repeat consistency and p50/p95 latency before comparing to the three other approaches. Ten images are a screening demonstration, not a reliability study.

TASK 2 | 5 MARKS

Levels of computer vision and image processing: compare pipeline depths

Low-level transforms pixels; mid-level extracts regions and attributes; high-level combines attributes into a grade.

2.1 Comparison of pipeline depth

Dimension	A. Low-level only	B. Low + mid-level	C. Low + mid + high-level
Stages	Blur / colour conversion / threshold / count dark pixels.	Adds mask/segmentation, contours, size, hue and spot features.	Adds classifier or decision policy over feature vector; emits grade + confidence.
Input → output	Image → binary image / dark-pixel count.	Image → segmented fruit, feature vector and rule-based grade.	Image/feature vector → class probabilities, grade and reason codes.
Shadow/stem errors	Dark stem end or shadow can be called defect.	Region location, shape, colour contrast and area can reject some false candidates.	Classifier may learn context when representative labels include shadows/stems; bad features still limit it.
Touching fruits / occlusion	No instance separation; dark pixels are all mixed.	Contours may merge touching fruit; occlusion produces biased area/length.	Cannot recover missing pixels without extra views/sensor evidence; should flag low-quality frames.
Lighting robustness	Low; fixed threshold moves with illumination.	Moderate if illumination normalized and colour is calibrated.	Potentially higher with training diversity but only within training domain.
Explainability / new grade	Easy threshold explanation; new grade means editing rules.	Measurements remain inspectable; need new features/rules.	Classifier needs labelled examples and confidence calibration; still needs known-failure logic.

2.2 Selected three-level pipeline and stage representations

Scene photons → raw camera samples → corrected BGR image [acquisition]
LOW: flat-field correction → mild denoise → colour conversion [image → image]
MID: fruit mask → contour / oriented box → mm dimensions [image → regions + measurements]
MID: calibrated hue/Lab statistics + defect components + size features [regions → feature vector]
HIGH: acceptance rules or compact SVM / random forest [features → class + confidence]
Decision: accept / reject / manual-review → timed output to ejector [decision → action]

Stage	OpenCV operations (examples)	Input	Output / level

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Radiometric correction	cv2.subtract / normalize; white-reference flat-field	Raw/dark/white reference frames	Corrected image; low-level
Noise/colour preprocessing	cv2.GaussianBlur; cv2.cvtColor	BGR image	Filtered BGR, gray, HSV or Lab image; low-level
Fruit segmentation	cv2.threshold / inRange, morphology, cv2.findContours	Corrected image	Binary mask / candidate regions; mid-level
Size / shape	cv2.contourArea, cv2.minAreaRect, calibration mapping	Fruit contour	Length, width, area in mm; mid-level
Defect candidates	Local contrast threshold, connected components, shape filtering	Fruit ROI / dark map	Component count, max diameter, inter-spot gap; mid-level
Decision	Threshold rules or cv2.ml.SVM / KNearest	Feature vector	Grade, confidence, reason codes; high-level

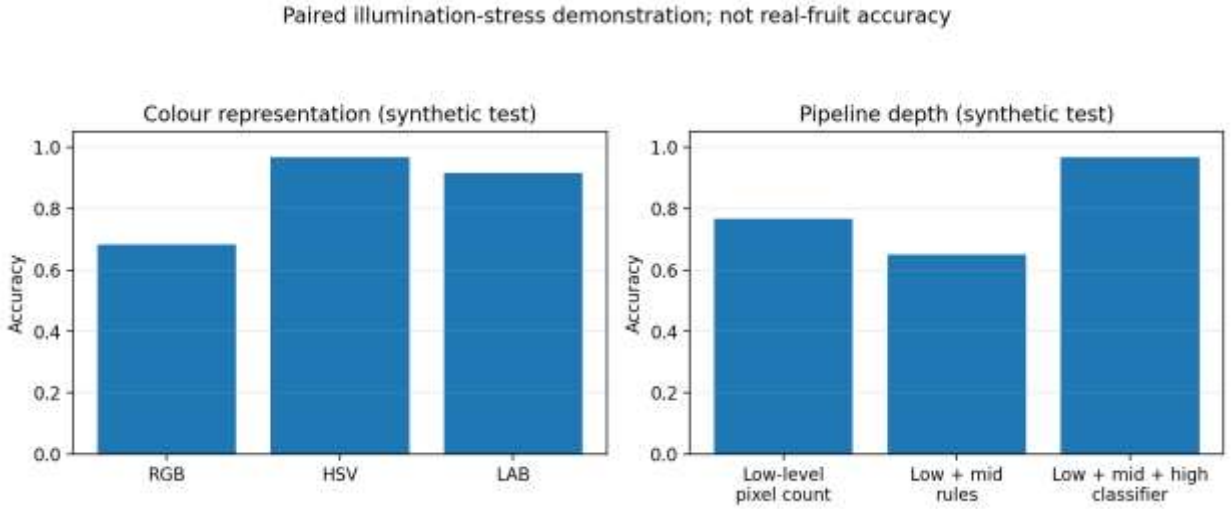


Figure 1. Executed controlled synthetic tests. These values are illustrative algorithm tests, not real-mango validation.

In the included synthetic run, the simple low-level pixel threshold achieved 76.67% accept/reject accuracy; the low+mid rule reached 65.00%; and a random-forest classifier trained only on reference-light synthetic images reached 96.67% on dim and bright variants of the same generated fruit identities. The run measured local mean processing times of 0.068 ms, 1.032 ms and 1.032 ms respectively for small 320×240 synthetic images on the current runtime. The paired same-fruit setup is not a true independent holdout; the times do not describe the edge PC and are not relevant to the required 1.2-second latency until rerun at production resolution on the target PC.

Synthetic pipeline	Test images	Measured accuracy	Local mean per-image time	Interpretation

Synthetic pipeline	Test images	Measured accuracy	Local mean per-image time	Interpretation
A. Low-level pixel threshold	60 dim/bright variants	76.67%	0.068 ms	Fast, but not enough context; misses / false alarms around dark objects and shadows.
B. Low+mid hand rules	60 dim/bright variants	65.00%	1.032 ms	Component features are useful, but the chosen fixed feature threshold was not robust in this synthetic case.
C. Low+mid+high random forest	60 dim/bright variants	96.67%	1.032 ms	Classifier combined available features; performance is optimistic because test images are paired variants of training identities.

The unexpected result that B scored below A is useful: a theoretically better pipeline can perform worse when a threshold, segmentation step or feature is poorly calibrated. Re-test with realistic camera images, independent fruit identities, blur, overlap, leaf occlusion and day-to-day variation.

2.3 CNNs and vision transformers compared with hand-crafted features

Approach	Representation learning	Data / compute implication	Placement in this design
Conventional CV	Engineer chooses edges, contours, colour statistics and component features.	Can start with tens to hundreds of calibration examples, but validation still needs representative large samples. CPU-friendly and easy to audit.	Best first deployment: calibrated geometry + inspectable features + compact classifier.
CNN	Early filters learn local edges/texture; deeper layers combine shapes and semantics.	Transfer learning can reduce new labels, but varied fruit/cultivar/lighting/defect data remains necessary. CPU inference cost depends on backbone and input resolution.	Possible high-level replacement after baseline and holdout protocol are stable.
Vision Transformer (ViT)	Attention mixes long-range spatial evidence; patch embeddings form learned representations.	Often higher input/compute demand; small-data fine-tuning may overfit without good pretraining/augmentation.	Evaluate only if it improves the independent holdout at required CPU latency.

Deep networks learn features corresponding roughly to low-, mid- and high-level visual representations within the model, but the pipeline no longer exposes each stage as a simple hand-labelled contour or feature value. For traceability, retain the original image, model hash, preprocessing version, confidence, and an explanation tool such as Grad-CAM/feature attribution as supporting evidence, not as proof by itself.

TASK 3 | 4 MARKS

Digital image fundamentals I: compare colour representations for ripeness

The colour space should preserve meaningful peel colour under lighting change and red blush while remaining easy to calibrate and explain.

3.1 Comparison of the three candidate representations

Question	A. RGB / BGR	B. HSV	C. CIE L*a*b*
What carries colour information?	Three camera channels; OpenCV stores BGR order by default.	Hue encodes dominant colour; S saturation and V brightness help mask low-colour background/highlights.	L* lightness; a* green↔red; b* blue↔yellow; compare with references using ΔE.
Brightness change	All channels scale/change with illumination and exposure.	Hue is often more stable for a moderate brightness scaling while S/V can gate unreliable pixels.	a*/b* tend to separate chromaticity; L* isolates lightness but is still illumination-dependent.
Red blush / mixed colour	Raw average can confuse same fruit with different red-patch fraction.	Single mean H can mix yellow/green and red hues; use circular hue stats and multiple region features.	a* responds to red blush; b* responds to yellow/blue; use distributions/percentiles, not only one average.
Hue wrap-around	Not applicable.	Hue is circular (OpenCV H is 0–179); values near 0 and 179 can be adjacent hues. Naive arithmetic mean is wrong near the wrap.	Not applicable.
Compute / explain	Simple and fast, but fixed channel thresholds are brittle.	Fast, intuitive hue rules; thresholds depend on camera/white balance.	Moderate conversion cost; ΔE / a*,b* comparisons can be calibrated to reference samples.
Choice	Retain raw BGR for evidence and conversion.	Good operational ripeness baseline after controlled lighting and hue-wrap handling.	Use as a second calibrated colour descriptor / audit cross-check, especially if Lab clusters are better on real held-out samples.

3.2 Executed synthetic colour-space comparison

The companion script generated 30 synthetic fruit identities (10 per ripeness class), each rendered at reference factor 1.0, dim factor 0.70 and bright factor 1.25. A nearest-centroid classifier was fitted on the 30 reference-level images and evaluated on the paired 60 dim/bright images. HSV features used a circular representation for hue; Lab used mean a*/b*-encoded channels (OpenCV output).

Representation	Dim accuracy (30)	Bright accuracy (30)	Combined accuracy (60)	Limit
RGB	60.00%	76.67%	68.33%	Synthetic controlled stress test only
HSV	96.67%	96.67%	96.67%	Synthetic controlled stress test only
LAB	86.67%	96.67%	91.67%	Synthetic controlled stress test only

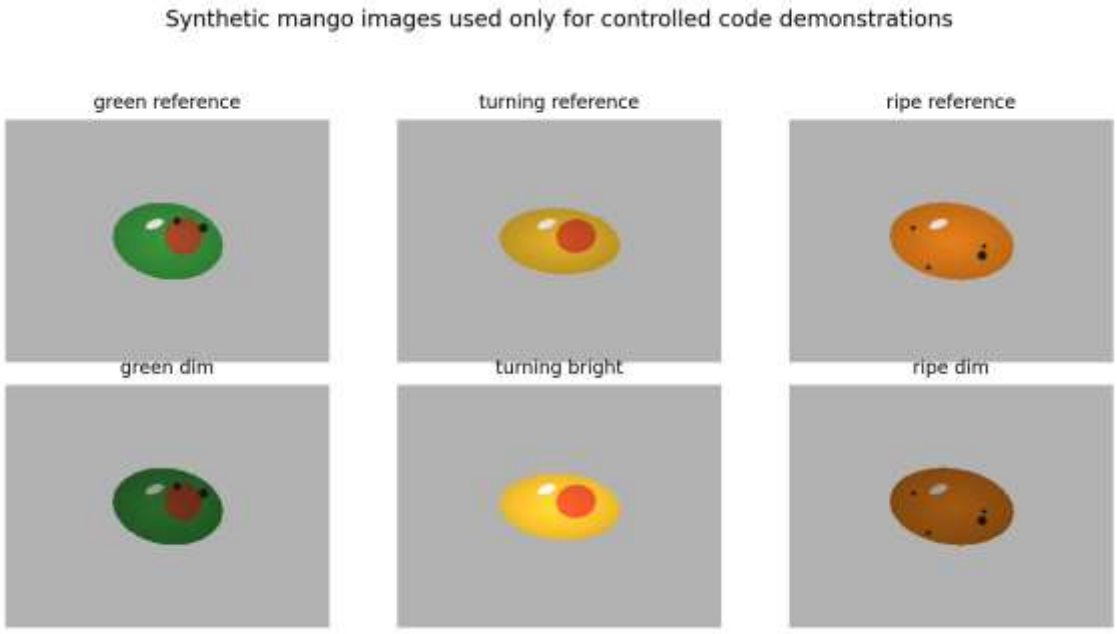


Figure 2. Synthetic examples at reference/dim/bright illumination; these are generated images, not camera photographs.

Result: HSV reached 96.67% combined accuracy, Lab reached 91.67%, and RGB reached 68.33% in this controlled synthetic setup. This result supports using HSV for a first rule-based feature, not a universal claim that HSV always beats Lab. The generator varies brightness by multiplication and creates simplified colours; real wax, cultivar, sensor white balance, skylight colour and local highlights can reverse the ranking. The same analysis must be rerun on at least 30 real fruits at two controlled light levels and on an independent test set.

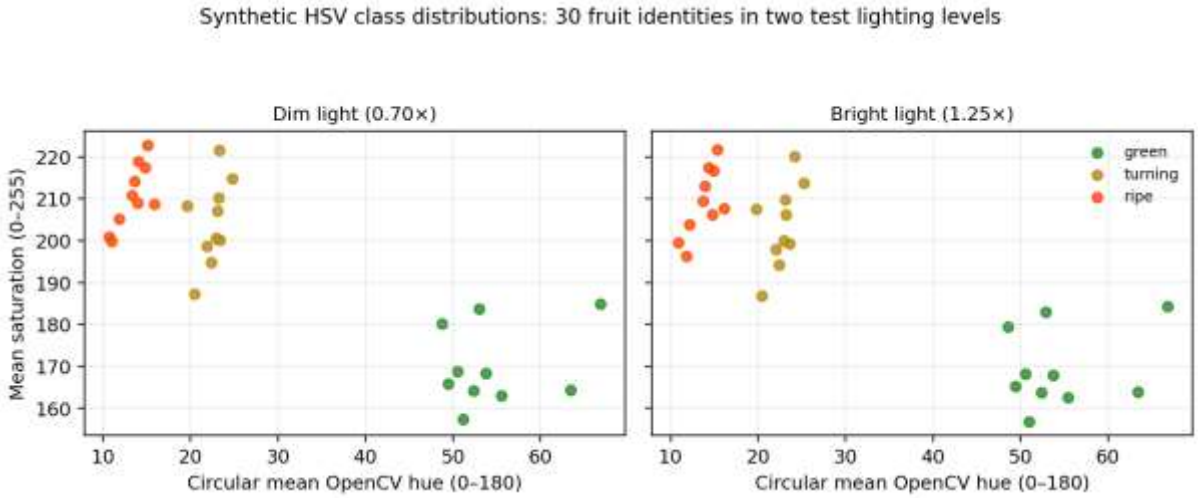


Figure 3. HSV hue/saturation distributions for 30 synthetic fruit identities under dim and bright conditions; real image distributions are required for production claims.

3.3 Image arrays, memory and uint8 arithmetic

A typical cropped image could have `img.shape = (960, 1280, 3)` and `img.dtype = uint8`. In NumPy, `img[row, column]` corresponds to `img[y, x]`; x is horizontal column and y is vertical row. Three 8-bit channels use k = 24 bits per pixel.

```
M = 960  # rows / image height
N = 1280 # columns / image width
k = 3 * 8 # bits per BGR pixel
b = M * N * k = 29,491,200 bits
bytes = b / 8 = 3,686,400 bytes ≈ 3.686 MB ≈ 3.52 MiB
```

For saturated addition, `cv2.add(uint8(250), uint8(10))` saturates to 255; NumPy `uint8(250) + uint8(10)` wraps modulo 256 to 4. Saturating addition is useful for display/processing when overflow should not wrap a bright value into a dark value.

```
a = np.array([250], dtype=np.uint8)
b = np.array([10], dtype=np.uint8)
cv2.add(a, b)  # array([255], dtype=uint8)
a + b          # array([4], dtype=uint8)
```

3.4 Consequences and pretrained network features

If daylight doubles while RGB thresholds are fixed, all three channels can brighten, highlights may clip and the same healthy peel can cross a defect/ripeness threshold. Auto white balance can change colour ratios too. If a BGR image is mistakenly interpreted as RGB, the red and blue channels are swapped; colour segmentation and Lab conversion become incorrect. Use a fixed exposure/white balance, diffuse controlled light, a colour reference patch, and explicit `cv2.COLOR_BGR2HSV` / `cv2.COLOR_BGR2LAB` conversions.

A pretrained ResNet/DINOv2 embedding can capture texture and shape information beyond hand-crafted colour. It is less directly interpretable than an a*/b* scatter plot or hue histogram and may respond to background/context. Compare a frozen embedding + compact classifier on the same independent 30+ fruit test set, record latency on the CPU-only edge PC, and show a counterexample set. Keep Lab/HSV measurements for buyer-audit explanations even if a network becomes the final high-level classifier.

TASK 4 | 4 MARKS

Digital image fundamentals II: compare pixel connectivity for defect counting

After thresholding candidate dark regions, pixel connectivity determines whether a diagonal streak becomes multiple spots or two spots merge.

4.1 Connectivity comparison

Behaviour	4-connectivity	8-connectivity	m-connectivity
Neighbour set	N4: up/down/left/right.	N8: N4 plus four diagonal neighbours.	N4 always; diagonal pixels connect only when the common 4-neighbours are background.
Diagonal two-pixel pattern	Two components.	One component.	One component when the two shared orthogonal neighbours are both background.
Sap-burn streak	Can split a diagonal streak into many one-pixel components.	Keeps diagonal strokes connected.	Reduces ambiguous diagonal paths but may still join corner-touching defects.

Behaviour	4-connectivity	8-connectivity	m-connectivity
Touching defects	May split real diagonal contact.	Can merge separate spots if their threshold masks touch diagonally.	Useful digital-topology compromise; not a universal separator for physically distinct diagonal-touching spots.
Noise	Isolated pixels count as components unless filtered.	Same issue; diagonal noise can join a valid spot.	Same issue; use minimum area, shape and local-contrast filters.
Implementation	OpenCV connectedComponentsWithStats(connectivity=4).	OpenCV connectedComponentsWithStats(connectivity=8).	Custom connected-component labeller is included in the companion script.

4.2 Executed binary-pattern check

Binary pattern	Foreground pixels	4-connected count	8-connected count	m-connected count	Observation
P1 horizontal pair	2	1	1	1	See scripted pattern generator
P2 diagonal pair	2	2	1	1	See scripted pattern generator

Binary pattern	Foreground pixels	4-connected count	8-connected count	m-connected count	Observation
P3 L triad	3	1	1	1	See scripted pattern generator
P4 diagonal chain	4	4	1	1	See scripted pattern generator
P5 two 2x2 blocks corner-touch	8	2	1	1	See scripted pattern generator

The test includes (1) horizontal pair, (2) diagonal pair, (3) L-shaped triad, (4) diagonal chain and (5) two 2×2 blocks touching at one diagonal corner. The diagonal pair produced 2/1/1 components under 4/8/m connectivity; the diagonal chain produced 4/1/1; the two diagonal-touching blocks produced 2/1/1. This confirms that 8- and m-connectivity can merge diagonal contacts. It does not establish which label count is "correct" for a real defect; that is a ground-truth annotation question.

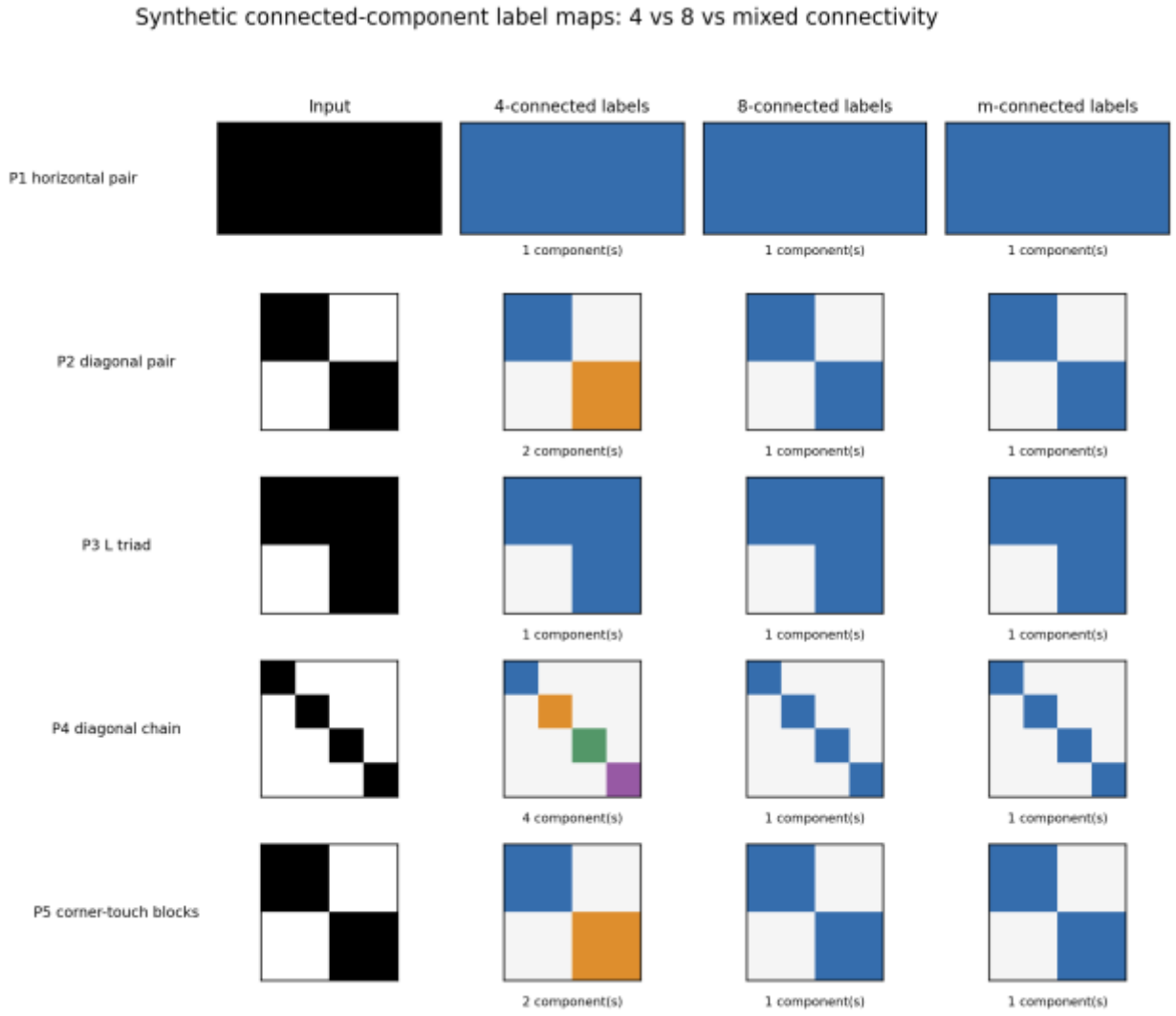


Figure 4. Synthetic binary inputs and connected-component label maps; this is not a thresholded fruit photograph.

4.3 Distance measures and 4 mm criterion

Distance	Formula for pixel displacement (dx,dy)	Use / limitation
Euclidean De	$\sqrt{dx^2 + dy^2}$	Direct straight-line distance; convert to mm after x/y calibration. Preferred for physical diameter/gap if the boundary estimate is valid.
City-block D4	$\lvert dx\rvert + \lvert dy\rvert$	Grid path distance; overestimates diagonal straight-line separation.
Chessboard D8	$\max(\lvert dx\rvert,\lvert dy\rvert)$	Minimum king-move count; underestimates Euclidean diagonal distance.

At 0.3125 mm/pixel, the buyer's 4 mm diameter threshold corresponds to 4 / 0.3125 = 12.8 pixels. Use calibrated maximum Feret diameter or an equivalent-circle diameter from the segmented lesion, and separately track measurement uncertainty. Example: for a 10×10 px diagonal bounding span, De is 14.14 px (4.42 mm), D4 is 20 px (6.25 mm), and D8 is 10 px (3.125 mm). Choosing D4 would overstate physical diagonal distance; D8 would understate it. Do not use city-block/chessboard distance as the sole physical diameter estimator.

Use watershed or instance segmentation only if real-image validation improves masks without violating CPU latency; either option still requires labelled fruit images.

TASK 5 | 5 MARKS

Image sensing and acquisition: compare sensor arrangements

The scene must be frozen sharply at 0.5 m/s, illumination must not flicker, and geometry must be synchronized with the conveyor.

5.1 Acquisition choices

Factor	A. Rolling-shutter area scan	B. Global-shutter area scan + strobe	C. Line-scan camera
Motion blur	Set by exposure time; rows capture at different instants.	All pixels expose together; a short LED pulse can freeze motion.	Each line is narrow in the motion direction; the encoder stacks lines into a 2-D image.
Skew / distortion	Rolling shutter can shear a moving fruit. If full-row readout spans 10 ms, belt displacement is 5 mm at 0.5 m/s. (10 ms is an example assumption.)	Negligible rolling-shutter skew; calibration still needed for lens/perspective.	Geometrically accurate only if line rate matches encoder speed; belt-speed drift stretches/compresses the image.
100 Hz lighting flicker	Risky with unsynchronized tube lights and variable exposure phase.	Strobe pulse synchronized to frame reduces ambient/flicker influence.	Line/strobe synchronization and stable DC lighting are required.
Colour	Bayer CFA reconstructs RGB by interpolation; a three-sensor prism camera separates channels better but costs more.	Bayer colour usually adequate after colour calibration.	Colour line-scan needs a colour sensor or multiple lines/sensors; moving fruit alignment complicates it.
Rate for baseline 0.3125 mm/px	Frame rate target at least 8 fps, with margin.	Choose ≥30 fps; 8 fps is the nominal presentation rate but extra frames support trigger reliability.	Line rate = 0.5 m/s / 0.0003125 m = 1,600 lines/s.
Integration/cost	Lowest initial cost, but less robust.	Recommended baseline; moderate integration.	Good for continuous high-speed surfaces but needs encoder, more integration and potentially multi-sensor colour.

5.2 Motion-blur calculation

Motion blur distance is v × exposure time. It is the physical displacement while the sensor is integrating light, before considering optical blur or readout artefacts.

Exposure	Blur at 0.5 m/s	Blur at 0.3125 mm/px	Effect on a 2 mm defect

Exposure	Blur at 0.5 m/s	Blur at 0.3125 mm/px	Effect on a 2 mm defect
0.1 ms	0.5 × 0.0001 = 0.05 mm	0.16 px	Negligible relative to the 6.4 px defect width.
1 ms	0.5 × 0.001 = 0.50 mm	1.60 px	Edge smear can materially reduce small-defect contrast.
4 ms	0.5 × 0.004 = 2.00 mm	6.40 px	Blur is as large as the defect; unreliable detection.

Use a short hardware-triggered pulse (about 50–100 µs as a design target), then adjust LED power/aperture/gain during commissioning. At 0.5 m/s, 100 µs means 0.05 mm blur. A longer exposure under tube lights can both smear a defect and integrate different phases of 100 Hz flicker.

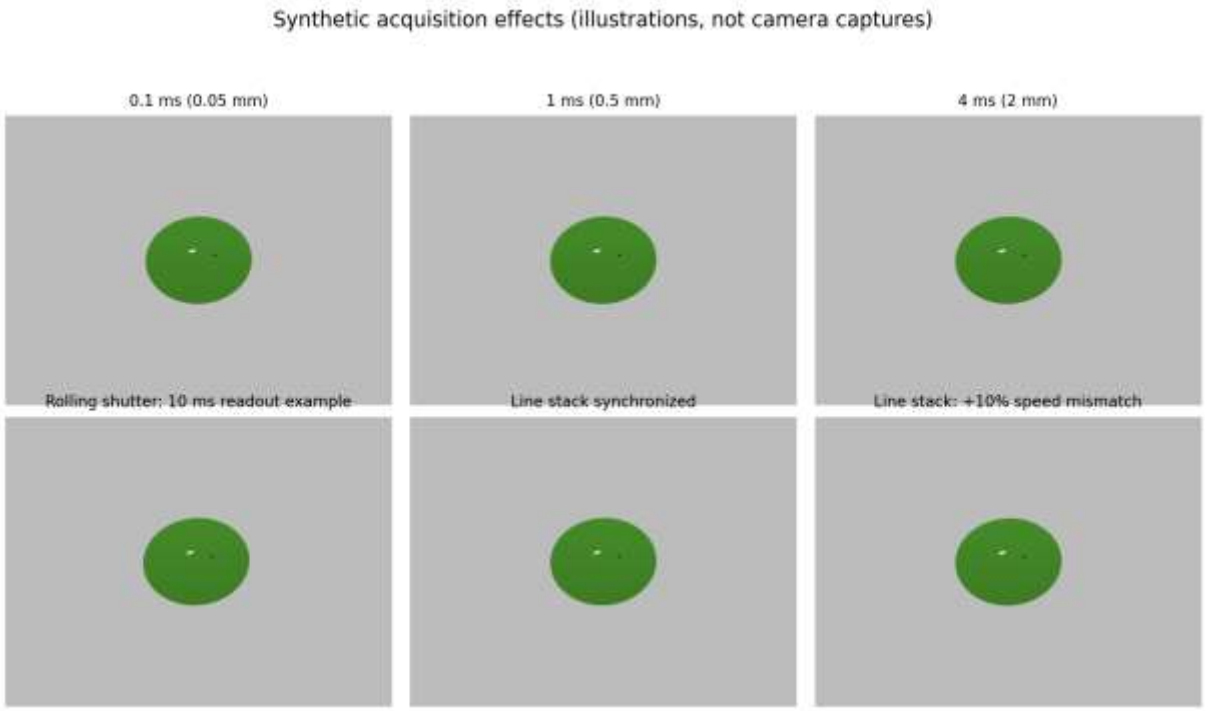


Figure 5. Software illustrations of motion blur, rolling-shutter shear and line-stack speed mismatch; no physical camera capture was supplied.

5.3 CMOS vs CCD and sensor conversion

Select a modern global-shutter CMOS sensor for the baseline because it can support high frame rates, hardware triggering, compact integration and suitable 8–12-bit output at less system cost than legacy CCD configurations. CCD can offer good uniformity in some applications, but it is not necessary for this line. Confirm the actual model's full-well capacity, read noise, effective dynamic range, bit depth and external-trigger latency from its datasheet.

Incident photons → photodiode electron/charge accumulation → charge-to-voltage conversion → analogue gain / correlated double sampling / analogue conditioning → ADC quantization → digital pixel code (e.g., 8-bit: 0–255; 12-bit: 0–4095) → packetized image → OpenCV array

5.4 Line rate and adverse cases

For 0.3125 mm per sample along belt motion, baseline line rate is 500 mm/s ÷ 0.3125 mm/line = 1,600 lines/s. A 300 mm field is 960 lines at this scale. The maximum 110 mm fruit occupies about 352 lines. If belt speed changes by 10% with no encoder synchronization, a fixed line rate causes about 10% length-scale error; use encoder pulses per mm and calibrate against a dimension standard.

A rolling-shutter camera with 4 ms exposure produces 2 mm motion blur at the original speed, enough to erase the smallest defect. Correct by lowering exposure and adding a strobe. Event cameras offer very high temporal resolution and encode brightness changes, but conventional colour and calibrated size/defect area are harder to obtain; a specialist event camera and new algorithms are required.
Multispectral/hyperspectral capture can reveal chemical/moisture differences invisible in RGB, but raises optics, illumination, calibration and data-rate cost. For the original visible peel task, the balanced RGB camera is more practical; Task 10’s NIR-only target changes that decision.

TASK 6 | 5 MARKS

Sampling: compare spatial resolutions

Sampling determines whether a 2 mm lesion spans enough pixels for reliable segmentation and whether image storage remains feasible.

6.1 Spatial resolution and 2 mm defects

Resolution	mm/px X	mm/px Y	2 mm pixels X	2 mm pixels Y	RGB8/frame	RGB8 @ 8 fps	7-day raw RGB8
640x480	0.625	0.625	3.2	3.2	0.922 MB	7.373 MB/s	1.858 TB
1280x1024	0.3125	0.29297	6.4	6.83	3.932 MB	31.457 MB/s	7.927 TB
2448x2048	0.1634	0.14648	12.24	13.65	15.041 MB	120.324 MB/s	30.322 TB

Calculations use the stated 400×300 mm FOV across the whole frame; they are geometric sample counts, not measured detector recall. The 1280×1024 case has different x/y scales because its aspect ratio is not the same as the 400×300 FOV. A 1280×960 crop gives 0.3125 mm/pixel in both directions. At 640×480, the 2 mm spot spans 3.2 pixels, only marginally above the practical 3–4 pixel target. At 1.3 MP it spans about 6–7 pixels, giving a stronger margin. The 5 MP configuration provides about 12–14 pixels across 2 mm.

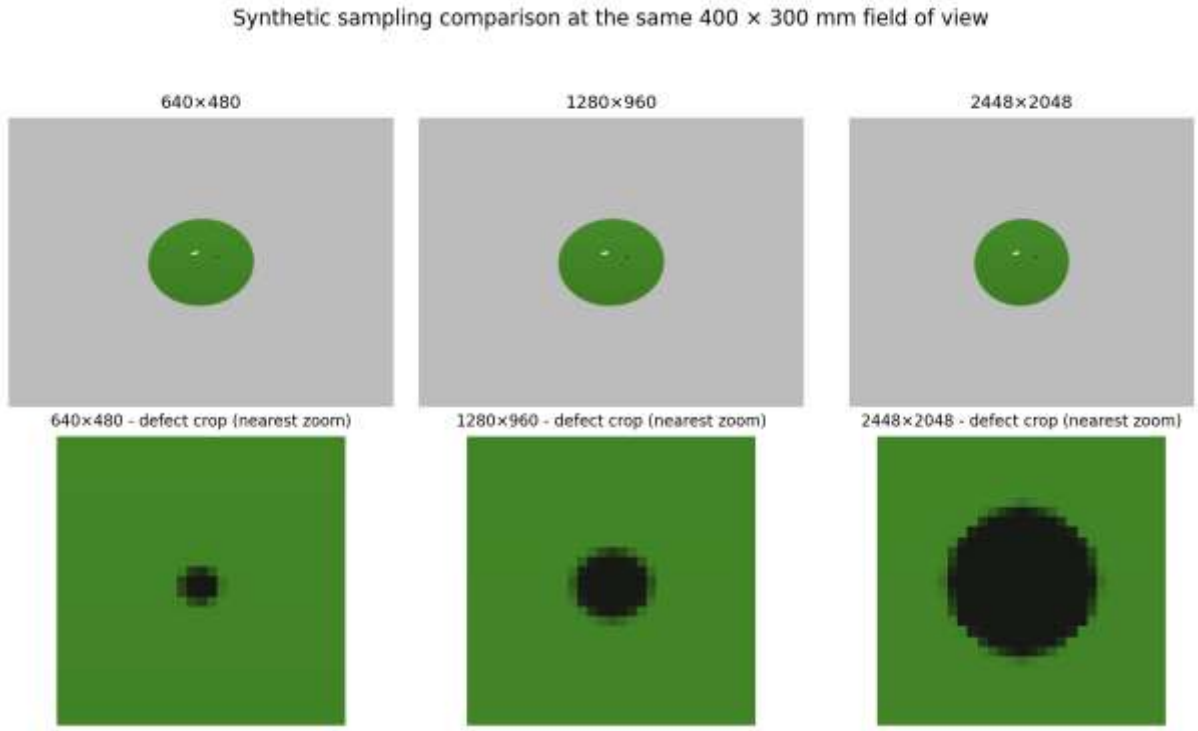


Figure 6. Re-sampled synthetic scene at equivalent 400×300 mm FOV. Detail crop is enlarged for viewing; it is not a real-camera defect-detection measurement.

6.2 Nyquist, dimensional error and aliasing

Resolution	Nyquist minimum (≥2 samples)	Practical 3–4 pixel test	Assessment
640×480	Passes nominal 2-pixel criterion: 3.2 px.	Marginal: 3.2 px; edge location changes by a large fraction of spot width.	Possible low-cost prototype; cannot claim ≤2% misses without real test data.
1280×1024	6.4–6.83 px.	Passes practical criterion.	Recommended baseline for 2 mm defects and ±2 mm size measurement, subject to calibration and contrast.
2448×2048	12.2–13.7 px.	Passes with margin.	Useful for smaller lesions but brings large data volumes and compute/network constraints.

If a contour edge is uncertain by ±1 pixel, the raw positional uncertainty is about ±0.625 mm at 640×480, ±0.31 mm at 1.3 MP and ±0.15–0.16 mm at 5 MP. This does not include perspective/fruit pose error, which can dominate. Nearest-neighbour downsampling can alias a small spot to zero or jagged boundaries; INTER_AREA is generally preferable when shrinking, although no resize method can restore detail that was never sampled.

6.3 Capture rate, seven-day storage and CPU throughput

For a fruit to fit completely inside a 300 mm longitudinal FOV when its length along the belt is 110 mm, the allowable frame interval is at most (300−110) mm / 500 mm/s = 0.38 s, or 2.63 fps, as a purely geometric minimum. To associate at least one distinct frame with every arriving fruit at 8 fruit/s, use at least 8 fps and provide margin with a ≥30 fps triggered camera. For Task 10 at 1 m/s, the geometric minimum is 1 / ((300−110)/1000) = 5.26 fps, but the per-fruit throughput target is 16 fps.

At 8 fruit/s for 10 hours/day, the archive contains 288,000 fruit images/day and 2,016,000 over seven days. Full-frame raw RGB8 would use about 1.858 TB/week at 640×480, 7.927 TB/week at 1280×1024 and 30.322 TB/week at 2448×2048. The storage device is 2 TB, so higher-resolution full frames require compression or region-of-interest storage. A 400×400 RGB8 crop is 480,000 bytes; retaining one such lossless uncompressed crop per fruit for seven days uses about 0.968 TB before metadata/filesystem overhead. That gives a workable baseline only if per-fruit crops rather than every full-FOV frame satisfy the audit requirement.

At the high-resolution setting, test the target 4-core CPU end-to-end at 8 fps, including acquisition, segmentation, inference, logging and image writing. The local synthetic benchmark is not evidence that the target PC can sustain full-resolution production capture. Use a bounded queue, avoid unneeded copies, process the fruit ROI, write images asynchronously, and trigger the ejector through deterministic hardware I/O rather than a slow UI callback.

6.4 Why super-resolution cannot rescue a missed defect

Real-ESRGAN or another super-resolution model can make a low-resolution image look sharper by using learned priors, but any texture not sampled by the sensor is generated/inferred, not newly measured evidence. If a 2 mm spot occupies 1–2 pixels or is eliminated by downsampling, upscaling cannot reliably

recover its true boundary, physical diameter or colour. For buyer audit, retain the original raw/cropped evidence; use enhanced images only as a visualization aid clearly marked as processed.

TASK 7 | 4 MARKS

Quantization: compare intensity resolutions

The bruise contrast is only 6–10 digital levels in an 8-bit image, so low-bit re-quantization can erase it.

7.1 Levels and expected bruise contrast

Representation	Levels	Quantization step expressed in 8-bit units	Effect on 6–10 level bruise contrast	Decision
12-bit raw	4,096	Camera-dependent mapping; ideally 1/16 of an 8-bit step if the same range is mapped over 12 bits.	May retain fine intensity differences if ADC dynamic range and noise support them; verify raw sensor data.	Prefer for acquisition if available, then maintain an auditable derived image.
8-bit	256	1 level.	6–10 levels survive as about 6–10 code values, provided contrast is not swamped by noise, glare or exposure change.	Recommended operational image depth for baseline if camera dynamic range is adequate.
6-bit	64	4 levels.	About 1.5–2.5 quantization bins; edge alignment and noise can reduce contrast.	Not recommended for archive without evidence.
5-bit	32	8 levels.	About 0.75–1.25 bins; a 6–10 level contrast can collapse to no or one code step depending on bin alignment.	Not sufficient for dependable bruise evidence.
4-bit	16	16 levels.	About 0.375–0.625 bins; contrast is mostly removed.	Reject for inspection and audit images.

The 12-bit statement is conditional: output code depth alone does not create extra sensor information if read noise, photon noise, clipping or illumination variation dominate. Inspect native linear/raw values and confirm effective number of bits. A 12-bit value stored in a 16-bit container often consumes two bytes/pixel unless packed; a re-quantized 4–6 bit image stored in a normal 8-bit PNG/JPEG may not reduce file storage at all unless a packed format is used.

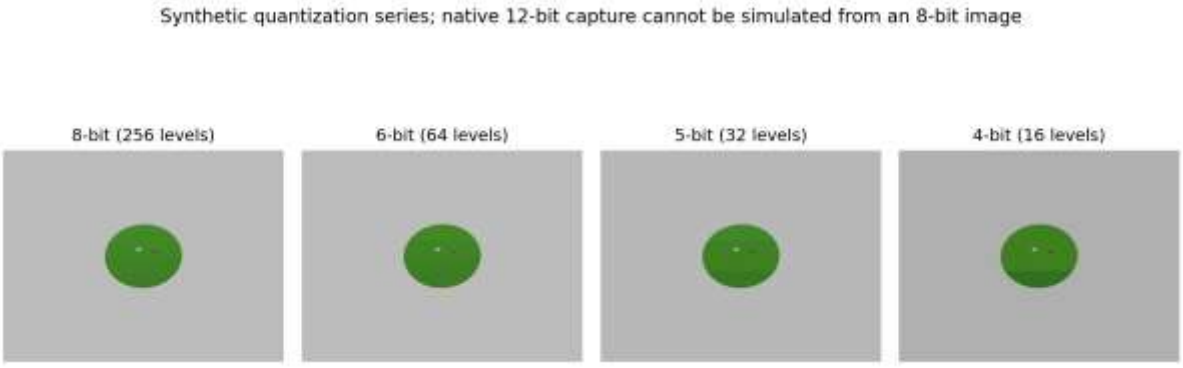


Figure 7. Same synthetic scene quantized to 8/6/5/4 bits. Upscaling this figure would not restore lost detail; a genuine 12-bit capture must originate in the sensor.

7.2 Storage and spatial-vs-intensity trade-off

For one 400×400 RGB crop per fruit at 8 bits/channel, uncompressed size is 400×400×3 = 480,000 bytes. At the baseline volume, seven-day storage is about 0.968 TB. A 16-bit-per-channel container doubles this to about 1.935 TB before metadata or filesystem overhead, leaving essentially no headroom on a nominal 2 TB device. Keep operational 8-bit colour crops plus selected native 10/12-bit raw samples or lossless raw ROIs where supported; size the archive after measuring actual compression on real fruit.

Under a fixed storage budget, do not sacrifice spatial detail for 5-bit quantization when the smallest defect is spatially small and bruise contrast is already only 6–10 levels. Spatial sampling ensures the lesion has multiple pixels; intensity depth preserves the per-pixel contrast. Choose a calibrated 8-bit operational stream with controlled lighting and retain raw 12-bit evidence selectively or use ROI/efficient lossless storage if a 12-bit audit trail is required.

7.3 Required quantization tests and bit depth vs model precision

Run the required quantization experiment in three parts: (1) produce 8-, 6-, 5- and 4-bit versions by reducing representable values, then record healthy-peel/bruise medians, contrast, lesion recall, false-positive rate, file bytes and contour bands at each tested light level; (2) review 5-bit images in the buyer-audit workflow and measure whether the bruise remains identifiable, because contrast may collapse; and (3) distinguish image bit depth from model precision. HDR acquisition captures a wider scene dynamic range before quantization, whereas INT8 model quantization reduces network weight/activation arithmetic. INT8 inference does not require storing the inspection image at 8-bit and must be validated independently.

TASK 8 | 5 MARKS

Image formation models: compare optics and lighting configurations

Geometry controls apparent size; photometry controls pixel brightness and apparent colour/defect contrast.

8.1 Geometric and photometric models

Pinhole projection (one axis): $x = f \cdot X / Z$

Photometric model: $f(x,y) = i(x,y) \cdot r(x,y)$

where i(x,y) is illumination and r(x,y) is surface reflectance (simplified model).

Factor	A. Ambient roof light + standard lens	B. Diffuse LED dome + standard lens + calibration	C. Diffuse LED dome + telecentric lens
Projection / size	Perspective size changes with object distance.	Perspective remains; checkerboard calibration and height correction reduce systematic error.	Near-orthographic magnification over specified telecentric depth range; reduces height-induced size variation.
Illumination	Daylight changes from 2,000 to 60,000 lux; tube lights flicker.	Controlled diffuse LED improves repeatability and reduces shadow/specular hotspots.	Same lighting benefit plus telecentric optics; illumination is still not automatically uniform.
Waxed glare	High risk from direct sources; specular pixels can clip.	Diffuse dome surrounds fruit; cross-polarization may help if light loss is acceptable.	Diffuse dome also needed; telecentric optics do not remove surface specularity.
Calibration	Lens distortion and changing light confound thresholds.	Calibrate camera intrinsics/distortion with checkerboard; map mm/pixel at relevant fruit plane.	High quality dimensional solution but large object-space telecentric lens is costly/heavy for 400 mm FOV.
Cost / choice	Lowest, not stable enough for export quality.	Recommended baseline.	Premium alternative if ±2 mm cannot otherwise be met and budget permits.

8.2 Focal length and perspective error

For a thin-lens/pinhole approximation, field of view is related to sensor width and working distance by:

focal length $f \approx$ (sensor width × working distance) / object-space field of view

For sensor width 6.8 mm, working distance 500 mm, FOV 400 mm:

$f \approx (6.8 \times 500) / 400 = 8.5$ mm.

The nominal lens choice is approximately 8.5 mm; a practical 8 mm lens may be selected and then moved/focused to match FOV, but exact calibration and lens distortion must be checked. The 500 mm

working-distance figure is only the camera-to-belt geometry; use sensor format and actual lens FOV calculations before purchase.

At the fruit top, camera distance is 410 mm rather than 500 mm. Magnification ratio = 500/410 = 1.2195, a 21.95% increase. A 100 mm edge measured at the top using a belt-plane scale could appear about 121.95 mm, a 21.95 mm bias, far beyond the ±2 mm requirement. Calibrate at the measurement plane, use height estimation plus a calibrated geometric model, or consider telecentric/stereo alternatives. A monocular depth network may help estimate relative height but must be measured against calibrated truth; it is not a replacement for calibration.

8.3 Separating illumination and reflectance

White-reference flat-field correction is the first choice: capture a uniform white target with the production lighting; compute per-pixel illumination gain; apply it to the raw fruit image after dark-offset correction. A simplified form is $I_{corrected} = (I-D)/(W-D) \times scale$, where D is a dark frame and W is the white reference. Avoid division by values near zero and mask saturated pixels. Homomorphic filtering is an alternative: take log intensity so $\log f = \log i + \log r$, suppress slowly varying illumination components and preserve higher-frequency reflectance/defect detail. Validate both with the same threshold and compare defect recall/false positives before and after correction.

Use a diffuser dome, fixed camera exposure/gain/white balance, a hood that blocks uncontrolled skylight, exposure-triggered strobe, and regular reference-card checks. A cloud-induced shadow can shift a fixed dark-pixel threshold; if the decision changes substantially after a small illumination change, that is a calibration/lighting failure, not a reason to hide the variation with aggressive enhancement.

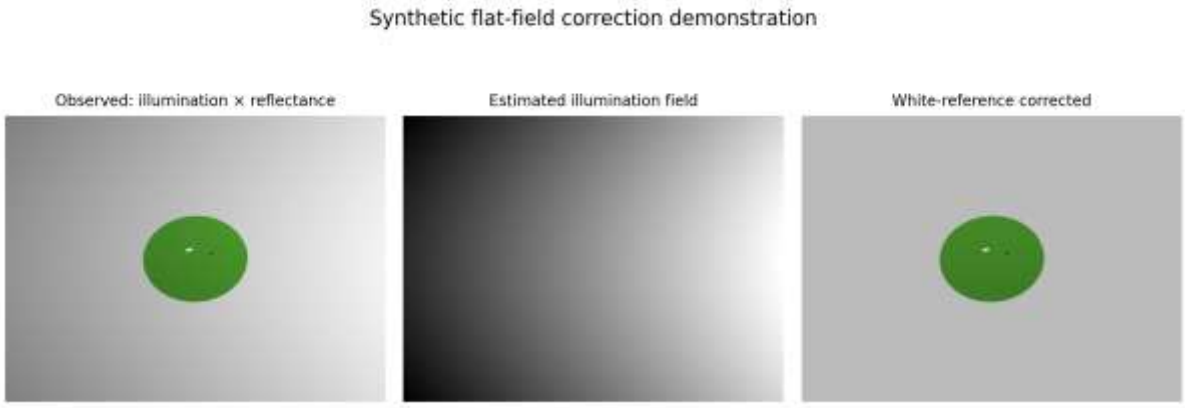


Figure 8. Simulated illumination-field correction illustrates the photometric model; validate on a white-reference image and real mango images.

8.4 Monocular depth vs telecentric imaging

Method	Strength	Limitation for ±2 mm size measurement	Validation test
Calibrated geometry + height correction	Physically interpretable; inexpensive if height can be measured.	Needs valid height estimate, known camera model and controlled pose.	Measure standard objects at belt and 90 mm height; compare length error over full FOV.

Method	Strength	Limitation for ±2 mm size measurement	Validation test
Monocular depth model	Can estimate relative depth from one image.	Depth is learned/ambiguous and depends on domain; absolute metric accuracy is not guaranteed.	Compare on held-out calibrated 3-D standards and mangoes; report MAE and 95th-percentile error.
Telecentric lens	Minimizes magnification variation over working range.	Large field object-space telecentric optics are expensive and have depth-of-field limits.	Verify magnification change across fruit-height range and full 400 mm field.

TASK 9 | 6 MARKS

Applications and integrated system: compare complete front-end configurations

Combines choices from Tasks 1–8; vendor prices are sourced where publicly listed, and unquoted items are explicitly treated as planning allowances.

9.1 End-to-end block diagram

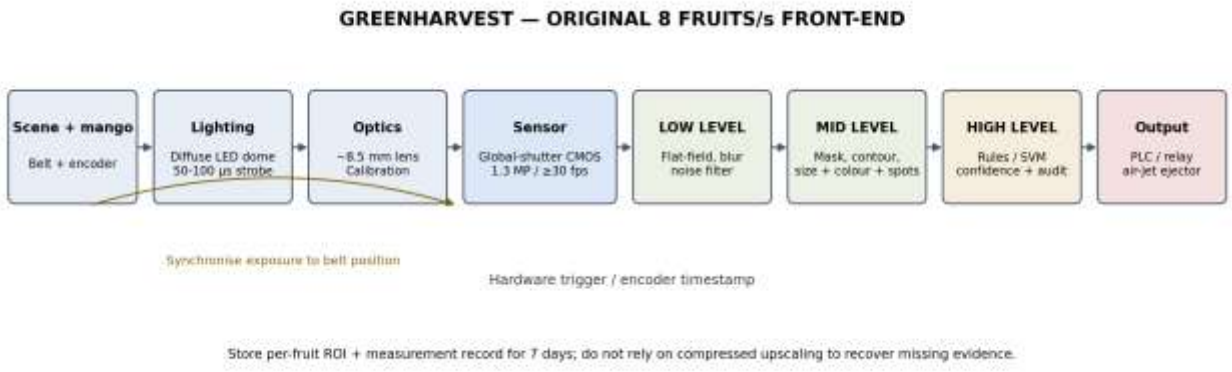


Figure 3. Proposed original-case front-end. Encoder-triggered illumination feeds low-, mid- and high-level processing and a timed ejector command.

9.2 Budget comparison

Public prices verified for this report include a Basler 1.3 MP GigE camera listed at ₹63,457 inclusive of taxes by Edmund Optics India [1], and a KUDAMM machine-vision dome light listed at ₹35,000 [2]. A 5 MP global-shutter 5GigE camera is listed at €869 [3]; the ECB's 27 October 2026 reference rate was €1 = ₹106.0635, implying about ₹92,200 before shipping/import charges/tax treatment [4]. Product price pages and exchange rates can change, and quotes should be reconfirmed before purchase.

Configuration	Hardware / software idea	Budget / price basis	Strength	Main limitation
A. Budget	0.3–1.3 MP rolling-shutter USB camera; ambient light; low+mid rules; compressed 8-bit JPEG.	Often lowest; a complete list must be quoted. Camera choice must be verified against motion specs.	Low initial cost and quick prototype.	Weak illumination/motion robustness; low confidence for 2% miss target.
B. Balanced (recommended for original case)	1.3 MP global-shutter GigE camera, diffuse LED dome, short strobe, calibrated standard lens, encoder, ROI audit images, low+mid+high pipeline.	Planning BOM below totals ₹1,50,000, but only camera/dome amounts are public listed prices; other items are allowances requiring quotes.	Best balance for original 2 mm defects and CPU-only deployment.	2% error limit is unproven; full-height perspective and physical fruit spacing need correction.

Configuration	Hardware / software idea	Budget / price basis	Strength	Main limitation
C. Premium	5 MP / line-scan, high-resolution lens, preferably higher bit depth; more data bandwidth.	5 MP 5GigE camera listed €869 (~₹92,200 at the cited reference rate, before import/taxes) [3,4]; lens/lighting/5GigE NIC are extra.	More samples per defect and better potential geometry.	Likely exceeds ₹1.5L; 5GigE interface is not a drop-in fit for the stated Gigabit-only PC.
D. Smart camera + onboard accelerator	Industrial smart camera with detector and digital output, external lighting/encoder.	Obtain a system quote; do not assume it fits ₹1.5L.	Integrated trigger, inference and I/O; lower host-PC workload.	Vendor-specific deployment, model retraining limits, price, storage/audit integration.

9.3 Candidate ₹1,50,000 BOM for Configuration B

Item	Budgeted INR	Price status / note
1.3 MP Basler acA1300-60gc camera	63,457	Public vendor listing; price inclusive of taxes on page [1]. Confirm exact shutter mode/trigger/bit-depth SKU.
Diffuse machine-vision dome light	35,000	Public marketplace listing for KUDAMM dome light [2]; verify diameter, brightness, and driver compatibility.
Fixed focal length C-mount lens (~8–8.5 mm)	12,000	Planning allowance; ask vendor for sensor coverage, distortion and resolution quote.
LED strobe driver / trigger interface	8,000	Planning allowance; compatibility with camera exposure output and dome light must be checked.
Encoder / trigger cable and interface	6,000	Planning allowance; required to synchronize to belt position.
Hood, mounts and baffles	8,000	Planning allowance; blocks skylight and supports repeatable geometry.
Calibration target / colour reference	1,500	Planning allowance; use certified dimensional target and stable colour patch if possible.
Industrial I/O / relay to ejector	5,000	Planning allowance; PLC/air-jet interface timing and electrical isolation to be verified.
GigE cable, power and connectors	4,000	Planning allowance.
Contingency	7,043	Remaining amount to reach ₹1,50,000; quotes or taxes can exceed this.

Item	Budgeted INR	Price status / note
TOTAL	150,000	Budget fits only if allowances and vendor quotation match. It excludes edge PC (given), installation labour and ongoing maintenance.

Budget caveat

The subtotal is a planning envelope rather than a purchase order. Because only two line items have cited public listings and several items are quote-based allowances, confirm GST, delivery, warranty, driver compatibility, lens field of view and mounting before claiming the system can be built for ₹1.5 lakh. A mismatch requires reducing scope or increasing budget, not silently substituting non-industrial parts.

9.4 Time budget and acceptance criteria

Stage	Target budget per fruit	Test / acceptance criterion
Trigger + exposure	≤5 ms including trigger handling (not exposure only)	Timestamp exposure and encoder count; reject duplicate/missed trigger frames.
Readout / transport / crop	≤20 ms target average, streaming with buffer	Sustain camera frame rate at the chosen ROI; queue depth must not grow over a full shift.
Low-level correction and segmentation	≤15 ms target	Run on target 4-core CPU at target resolution; capture worst-case shadows/glare.
Mid-level measurement/features	≤15 ms target	Validate dimensional error against calibrated targets and defect count on labelled data.
High-level decision + log	≤10 ms target	Record grade, confidence, image ID and reason code; fail-safe on timeout/poor image.
Output / safety margin	≤10 ms target	Measure camera-to-ejector end-to-end latency and verify actuator timing; do not rely solely on software timestamp.
Pipeline total target	~75 ms/fruit processing budget, with parallel stages	At 8 fruit/s the arrival interval is 125 ms; throughput, not only single-frame latency, must be sustained. The physical 1.2 s travel time gives the gross decision deadline.

These numbers are design targets, not measured latencies. The system should complete decisions well before the fruit reaches the ejector and continuously sustain the target arrival rate. At 8 fruit/s, one decision is expected every 125 ms; a pipelined CPU can have latency longer than 125 ms and still sustain throughput, but it must remain below the 1.2 s transport deadline. Confirm hardware triggering and output timing end to end.

9.5 Design decision table

Decision	Justification / evidence	What if not?
Global-shutter CMOS	0.1 ms pulse gives 0.05 mm blur at 0.5 m/s; avoids rolling-shutter skew.	Long exposure can smear 2 mm defects over their own width; rolling shutter can shear size.
Diffuse dome + hood + short strobe	Reduces reflections/shadows and daylight sensitivity; camera-triggered timing bypasses 100 Hz flicker variation.	Thresholds drift under 2,000–60,000 lux roof changes; glossy highlights hide defects.
Calibrated lens + height correction	8.5 mm nominal focal-length estimate; fruit-top perspective scale can change about 21.95%.	±2 mm size requirement is not credible with uncorrected belt-plane scale.
1.3 MP for Task 9, ROI storage	About 6–7 pixels across 2 mm; 400×400 RGB8 per fruit uses ~0.968 TB/week.	640×480 is marginal; full 1.3MP full-FOV images exceed 2 TB/week at the required rate.
Three-level pipeline + explicit review reject	Combines measurable size/colour/spot features and final decision; uncertainty is recorded.	Low-level-only dark threshold falsely rejects shadows/stems and misses unfamiliar lesions.
Versioned images + labels + model	Supports audits, retraining and root-cause analysis.	Without source images and version IDs, buyer disputes and performance drift cannot be reconstructed.

9.6 Transfer to three other computer-vision applications

Application	Sensing band	Spatial sampling	Quantization	Image formation vs mango
Medical X-ray analysis	X-ray attenuation; often grayscale detector.	Chosen by anatomy and diagnosis; high resolution for small structures, not a fixed conveyor FOV.	Often 12–16 bit or higher detector codes to preserve attenuation differences.	Transmitted intensity through anatomy; contrast depends on tissue attenuation/scatter, not peel reflectance.
Satellite remote sensing	Visible, NIR, SWIR, thermal or multi/hyperspectral bands.	Ground sampling distance can range from sub-metre to tens of metres depending on sensor/mission.	Commonly 10–16-bit radiometric values; calibrated radiance/reflectance products.	Sun illumination, atmosphere, surface reflectance, viewing geometry and terrain; illumination corrections are essential.

Application	Sensing band	Spatial sampling	Quantization	Image formation vs mango
Vehicle number-plate recognition	Visible RGB plus near-infrared illumination / NIR camera at night.	Enough pixels across plate and characters; motion and perspective dictate lens/resolution.	8–12 bit often sufficient with controlled exposure; HDR may help headlights.	Perspective projection + reflectance/retroreflection, motion blur, glare and changing backgrounds.

9.7 Test design and theoretical counterexample

Do not test only clean, evenly lit images. Include cloudy/daylight changes, gloss, stem ends, leaf occlusion, touching fruit, partly hidden spots, conveyor vibration, weak bruises, wet fruit, dust on lens, blur, and fruit at different heights. A theoretical expectation that the low+mid pipeline beats low-level-only can be contradicted by the synthetic run: low-level scored 76.67%, whereas the fixed low+mid rule scored 65.00%. The measured cause is not a universal algorithm fact; the synthetic feature extractor's threshold/component behaviour was poorly matched to the generated data. This is a useful controlled counterexample, but real fruit is required to identify the production cause.

For the buyer's 2% limit, report confusion matrix, false-accept (defective fruit accepted), false-reject (good fruit rejected), precision, recall, class-wise F1 and confidence intervals. A rough statistical guide: if zero misses are observed in about 149 independent defect-positive test cases, the one-sided 95% upper bound on a defect-miss probability is approximately 2%. This does not replace a formal sampling plan; correlated fruit/batches or a nonrepresentative defect mix weakens the claim.

TASK 10 | 8 MARKS

Re-apply and adapt the imaging front-end: 0.5 mm defects, NIR and 16 fruit/s

New buyer: 0.5 mm fruit-fly stings and internal bruises, some visible only in NIR; belt speed doubles, daylight variation worsens, the same PC and ₹1,50,000 budget remain.

10.1 Original assumptions that no longer hold

Original assumption	Why it fails in Task 10	Required change
Smallest defect = 2 mm	Target is now 0.5 mm, four times smaller.	Increase sampling to ≤0.167 mm/px (3 samples) and preferably ≤0.125 mm/px (4 samples).
Belt speed 0.5 m/s	Speed is 1.0 m/s; exposure blur doubles.	Use shorter strobe pulse, hardware encoder trigger and recalibrated motion model.
8 fruit/s	Target is 16 fruit/s, 62.5 ms inter-arrival interval.	Increase acquisition and processing throughput; check physical pitch against fruit length.
Shed daylight 2,000–60,000 lux	Skylights make variation still stronger.	Enclose hood, control illumination, monitor saturation and use fixed exposure/white balance.
Visible colour only	Some internal bruises/stings are visible only in NIR.	Use a validated NIR-sensitive channel, suitable wavelength-specific illumination and visible/NIR calibration.
1.2 s ejector deadline	New distance travel time is 0.6 m / 1.0 m/s = 0.6 s.	Decision must be complete before arrival; use deterministic pipelining and a safety margin.
2 TB / 8 fruit/s	16 fruit/s doubles image count; 7-day full-frame storage doubles.	Crop per fruit, measure actual lossless compression, and/or negotiate storage/retention/network changes.

10.2 Recalculated sampling, blur, line rate and data rate

For a 0.5 mm defect to span 3–4 pixels, the ground sampling interval should be 0.5/3 = 0.167 mm/pixel to 0.5/4 = 0.125 mm/pixel. Across 400 mm this requires about 2,400–3,200 pixels horizontally; across 300 mm it requires 1,800–2,400 pixels vertically. A 2448×2048 (about 5 MP) sensor over the stated FOV yields 400/2448 = 0.1634 mm/pixel horizontally and 300/2048 = 0.1465 mm/pixel vertically, so a 0.5 mm feature spans about 3.06×3.41 pixels. This is just around the practical lower bound; optics, focus, contrast and demosaicing can reduce effective detail.

Condition / resolution	Calculation	Result / decision
0.5 mm spatial sampling	3–4 px across: GSD 0.167–0.125 mm/px.	Aim for 0.125–0.167 mm/px and validate modulation/contrast in real NIR images.

Condition / resolution	Calculation	Result / decision
5 MP over 400×300 mm	0.1634 mm/px X; 0.1465 mm/px Y.	3.06 px × 3.41 px across 0.5 mm; borderline-acceptable only if the optic and strobe preserve edge contrast.
Blur at 1.0 m/s, 0.1 ms	1.0 × 0.0001 m = 0.1 mm.	About 0.61–0.68 pixel at 5 MP sampling; use 50–100 µs and verify available light.
Blur at 1.0 m/s, 1 ms / 4 ms	1 mm / 4 mm.	About 6–7 px / 24–27 px at 5 MP; unacceptable for 0.5 mm defects.
Line-scan rate at 0.1634 mm/line	1000 mm/s ÷ 0.1634 mm = 6,120 lines/s.	Use at least about 6.2 kline/s at horizontal scale; for 0.125 mm/line target 8,000 lines/s.
5 MP RGB8 at 16 fps	2448 × 2048 × 3 bytes × 16 = 240.65 MB/s ≈ 1.93 Gbit/s.	Exceeds a practical 1 Gigabit Ethernet payload and the PC's stated link; use ROI/reduced stream, faster interface/NIC, or different architecture.
5 MP mono 8-bit at 16 fps	2448 × 2048 × 1 byte × 16 = 80.2 MB/s (~642 Mbit/s).	Can approach GigE payload feasibility, but this is not a colour ripeness image; 12-bit packed is about 120.3 MB/s before overhead.
Archive at 16 fruit/s	4,032,000 fruit images in seven 10-hour days. A 700×700 RGB8 crop is 1.47 MB each, about 5.93 TB/week uncompressed.	A 2 TB archive cannot store every such uncompressed ROI for seven days. Measure lossless compression or negotiate retention/storage/data format.

10.3 Strategies A, B and C

Strategy	0.5 mm surface defect	NIR-only defect	Time/storage/cost	Assessment
A. Retain camera + ROI	Original 1.3 MP sampling ≈0.31 mm/px gives only ~1.6 px across 0.5 mm. ROI reduces compute but not sensor sampling.	Cannot detect a defect not present in visible RGB evidence.	Lowest incremental cost and easiest integration.	REJECT as final solution; retain only as a temporary process monitor.

Strategy	0.5 mm surface defect	NIR-only defect	Time/storage/cost	Assessment
B. Higher-resolution global shutter + short strobe + hood + 10/12 bit	5 MP can give ~3.1–3.4 px across 0.5 mm; a 50–100 µs pulse controls blur.	Only if the selected sensor/optics are actually NIR-sensitive and NIR illumination is added; a standard colour camera alone is not proof of NIR capability.	High data rate; a 5 MP 5GigE model is listed at €869 before shipping/import/tax; PC interface is GigE.	Necessary modification for sampling and motion, but visible-only version fails NIR-only requirement.
C. RGB + NIR line-scan or two-camera design	Can use ≥0.125–0.167 mm/px and encoder-driven 6,000–8,000 lines/s or synchronized 5 MP frames.	Directly captures a calibrated NIR channel using wavelength-specific illumination.	Highest integration/data/storage cost; needs cross-channel registration, separate exposures/sensors and possibly faster network/storage.	Best technical path, but the full set of constraints is not proven feasible within ₹1.5L and the stated PC.

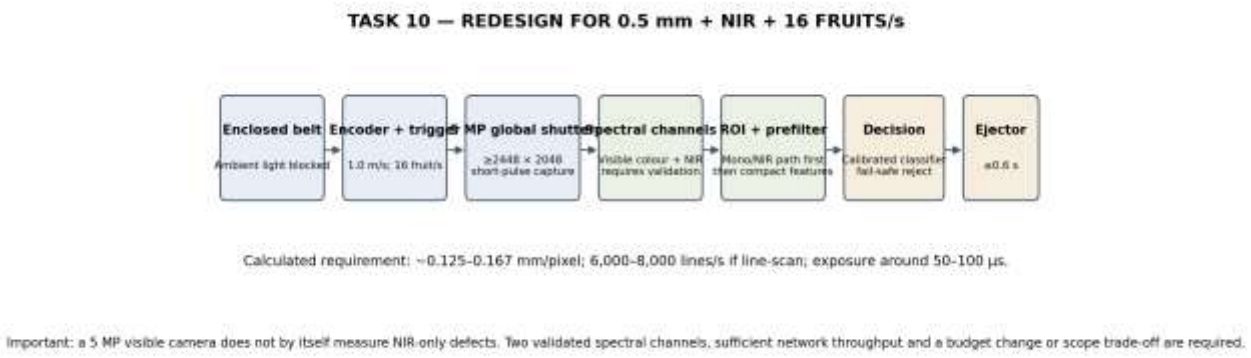


Figure 4. Task 10 redesign concept. This diagram is a requirements architecture, not a claim that all components fit the present budget.

10.4 Final RETAIN / MODIFY / REDESIGN decision

Decision: REDESIGN (with a cost/network feasibility gate)

Do not retain the Task 9 camera as the final inspector. A 5 MP-class global-shutter sensor, short pulse lighting, enclosed hood and encoder synchronization are required for the 0.5 mm target. NIR-specific evidence requires a validated NIR-sensitive sensor/channel and NIR illuminator. The cited 5 MP candidate is 5GigE, while the provided PC has Gigabit Ethernet; a standard RGB camera does not by itself prove NIR capability. The same ₹1,50,000 cap and 2 TB archive therefore do not support a defensible guarantee of all Task 10 requirements without a vendor-quoted, measured redesign. If the cap cannot change, pilot one priority (surface-visible 0.5 mm inspection or NIR-only internal bruise

inspection) and record which requirement is deferred; do not claim full compliance.

Components that remain suitable: the 4-core CPU can perform classical preprocessing and compact feature calculations; Python/OpenCV/NumPy remains appropriate; encoder timestamps, calibration targets, ROI processing, per-fruit metadata, controlled lighting and human review remain valid. The missing capability is not solved by a better software classifier alone: sampling, spectral sensitivity, throughput, network transport and audit capacity all change.

10.5 Few-shot learning for new NIR-specific defects

A pretrained foundation model can reduce the number of labels needed to start by using embeddings and a small set of labelled examples or by adapting a pretrained detector. But a model pretrained mainly on ordinary RGB photos may have poor transfer to NIR intensity/texture, camera response, wavelength, exposure and cultivar. A few examples can overfit, hide the true defect distribution and create unacceptable false acceptance. Treat few-shot adaptation as a way to propose a candidate model, not as evidence that a new defect is safe.

1. Collect NIR examples across independent fruit, defect severity, fruit age, cultivar, moisture, depth, speed and illumination. Keep lot/date-level splits so nearly identical fruits do not leak across train/test.
2. Start with a small annotated pilot, then grow the dataset using active learning on uncertain/novel cases. Have two trained reviewers adjudicate defect masks and retain disagreement records.
3. Freeze sensor, wavelength, lens, strobe, normalization, model weights, prompts, library versions, training dataset checksum and thresholds. Store original NIR/RGB inputs and prediction metadata for every audited fruit.
4. Report defect-specific false-accept rate, recall, false-reject rate, calibration, p50/p95 latency and confidence intervals on a holdout set. Revalidate after any sensor, lighting, preprocessing, camera firmware or model change.
5. Use an uncertainty/novelty gate and route out-of-distribution fruit to manual review. A high confidence score must not be interpreted as validated probability unless calibrated on the target domain.

Appendix A - Code execution and experimental protocol

The companion package contains `mango_cv_assignment.py`, a 90-image synthetic dataset, CSV outputs and diagrams. The code is runnable on Windows, Linux or macOS after installing Python, OpenCV, NumPy, pandas and scikit-learn. Synthetic images are intentionally marked as synthetic.

A.1 Run the companion experiment

```
python -m pip install opencv-python numpy pandas scikit-learn
python mango_cv_assignment.py --out results --dataset synthetic_dataset
```

The script creates/updates the results folder with:

Output	Purpose
colour_space_synthetic_results.csv	RGB / HSV / Lab paired illumination test and accuracy by dim/bright condition.
pipeline_depth_synthetic_results.csv	Low-, low+mid- and low+mid+high-level synthetic decision accuracy and local time.
pipeline_depth_per_image.csv	Per-image synthetic predictions and timing records.
connectivity_synthetic_results.csv	Component counts/areas for five synthetic binary patterns using 4-, 8- and m-connectivity.
sampling_storage_calculations.csv	mm/pixel, defect samples, RGB8 data rate and seven-day full-frame storage.
quantization_calculations.csv	Quantization-step versus 6/8/10-level contrast calculations.
manual_ratings_template.csv	Blank 30-fruit × 3-evaluator × 2-round form; fill to calculate kappa.
run_summary.json	Compact machine-readable result summary and warnings.

A.2 Using actual mango images

Create `real_mango_images/` and place one-fruit-per-image files there. Use fixed camera settings and include a scale/calibration target in commissioning shots. Run:

```
python mango_cv_assignment.py --out results --dataset synthetic_dataset --real-dir real_mango_images
```

This produces `real_image_feature_measurements.csv` with image dimensions, simple fruit mask area, colour statistics, dark-component count and local runtime. It intentionally does not claim accuracy because a list of images without verified labels cannot establish true positives, false positives or dimensional error. To finish the actual experimental portion, annotate images, add the labels and independently measure reference size/spot count, then compare system output to adjudicated truth.

A.3 Example core operations used

```
img = cv2.imread(image_path)  # OpenCV order: B, G, R
hsv = cv2.cvtColor(img, cv2.COLOR_BGR2HSV)
lab = cv2.cvtColor(img, cv2.COLOR_BGR2LAB)
mask = cv2.inRange(hsv, lower_hsv, upper_hsv)
mask = cv2.morphologyEx(mask, cv2.MORPH_CLOSE, kernel)
contours, _ = cv2.findContours(mask, cv2.RETR_EXTERNAL, cv2.CHAIN_APPROX_SIMPLE)
```

```
area_px = cv2.contourArea(max(contours, key=cv2.contourArea))
rect = cv2.minAreaRect(max(contours, key=cv2.contourArea))
labels, stats, centroids = cv2.connectedComponentsWithStats(defect_mask, connectivity=8)[1:]
```

For actual calibrated length, transform pixel coordinates through the calibrated image-to-plane mapping; do not simply multiply every contour dimension by a guessed scale if perspective and fruit height vary. For real threshold selection, freeze parameters before holdout evaluation and record them in the audit metadata.

A.4 Required experiment log

Experiment	Record at minimum	Pass / fail gate
Manual / image-assisted	Fruit ID, evaluator, round, label, defect count, size, decision, time.	Pairwise/intra-rater κ and per-fruit disagreement list.
Colour representation	Image ID, light level, class, RGB/HSV/Lab distributions, accuracy, confusion matrix.	Choose by independent holdout accuracy and illumination robustness; include red-blush cases.
Connectivity	Binary mask, reference spots, 4/8/m counts, diameters, gaps, run time.	Correct spot count and size threshold on ten real images plus synthetic fixtures.
Acquisition / sampling	Exposure, strobe, encoder speed, resolution, trigger count, blur, line/frame rate.	No missed trigger; defect occupies ≥3 pixels preferably 4; dimensions within ±2 mm.
Quantization	Native code values, contrast healthy vs bruise, bit depth, file bytes, detection rate.	No clinically/contractually meaningful contrast loss; original evidence retained.
Integrated pipeline	Ground truth, grade, confidence, feature reasons, per-stage timings, latency, queue depth, storage/day.	Independent test error and defect-miss rate within buyer limit with confidence interval.
Task 10 spectral redesign	RGB and NIR capture, wavelength, registration error, line/frame rate, network bytes/s, 0.5 mm targets.	NIR-only defects detectable; <0.6 s decision path; data budget and hardware sustained for full shift.

Appendix B - Key equations and data-quality checklist

B.1 Equations used

Quantity	Equation	Applied result / note
Belt pitch	pitch = speed / fruit rate	0.5 m/s ÷ 8/s = 62.5 mm; inconsistent with 110 mm longitudinal fruit.
Motion blur	blur_mm = speed_mm_s × exposure_s	At 0.5 m/s, 0.1/1/4 ms gives 0.05/0.5/2 mm.
Spatial scale	mm/px = FOV_mm / pixel_count	400/1280 = 0.3125 mm/px; 2 mm spans 6.4 px.
Pixels across target	target_mm / (mm/px)	0.5 mm at 0.1634 mm/px is ~3.06 px.
Line rate	speed_mm_s / desired_mm_per_line	0.5 m/s at 0.3125 mm/line = 1,600 lines/s; 1.0 m/s at 0.125 mm/line = 8,000 lines/s.
Raw frame bytes	width × height × channels × bytes_per_channel	RGB8: 3 bytes/pixel; 12-bit packed mono is about 1.5 bytes/pixel, while uint16 containers use 2 bytes/pixel.
Archive size	image_bytes × images_per_day × retention_days	Per-fruit ROI and measured compression determine if 2 TB is enough.
Pinhole magnification ratio	Z_belt / Z_top	500/410 = 1.2195; uncorrected top-plane scale bias about +21.95%.
Cohen κ	(p_o − p_e) / (1 − p_e)	Use for nominal human labels; interpret with prevalence and disagreement counts.
Zero-miss upper bound	p_upper ≈ 1 − 0.05^(1/n) for one-sided 95%	n≈149 independent defective cases with zero misses gives an upper bound near 2%.

B.2 Before sign-off

- ☐ Confirm the fruit-spacing contradiction is resolved in the physical line layout.
- ☐ Calibrate lens distortion and pixel-to-mm mapping across the full field and fruit-height range.
- ☐ Fix and record exposure, gain, white balance, lighting current, camera firmware and calibration version.
- ☐ Use strobe/encoder triggering and verify the real 0.6 m ejector distance with timestamps.
- ☐ Collect real images across defect types, fruit sizes, cultivars, gloss, shadows, occlusion and shifts.
- ☐ Use independent lot/date holdout validation; do not split near-duplicate frames across training and test.
- ☐ Report false-accept rate and the buyer's actual metric separately from overall accuracy.
- ☐ Measure 4-core CPU latency/throughput, queue behaviour, NIC throughput and image-write throughput for a full-shift test.
- ☐ Measure actual lossless ROI size and prove seven-day retention with safety headroom.

- ☐ For Task 10, validate the NIR channel and show NIR-only defects in raw evidence; do not infer them from RGB.
- ☐ Archive original image ROI, grade, confidence, model/code hash, feature reasons, calibration ID and timestamp.
- ☐ Keep a manual-review route for low-confidence, saturated, blurred, occluded or novel fruit.

References and source notes

Vendor prices and product specifications below were checked on 29 October 2026. Listed prices are point-in-time web listings, not purchase quotes. Currency conversion is shown only to estimate the premium camera magnitude; import duties, shipping and tax treatment may differ.

1. [1] Edmund Optics India. Basler ace acA1300-60gc Color GigE Camera, CS Mount. Listed price ₹63,457 inclusive of taxes; 1.3 MP, 1280×1024, 60 fps listing. Open source page
2. [2] Aajjo / KUDAMM Corporation. Machine Vision Dome Light. Listed price ₹35,000; verify emitted size, brightness and driver compatibility with supplier. Open source page
3. [3] Alpha TechSys / Basler. Basler a2A2448-105g5cBAS, 5 MP global-shutter 5GigE camera. Listed net price €869; 2448×2048, 106 fps; quote and availability subject to vendor confirmation. Open source page
4. [4] European Central Bank. Euro reference exchange rates, 27 October 2026: EUR 1 = INR 106.0635. Used only for approximate comparison; actual settlement rate may differ. Open source page
5. [5] Basler. acA1300-60gmNIR camera documentation. Visible + NIR spectrum 0.4–1.1 µm, 1.3 MP; the cited page is a specification reference, not a local price quote. Open source page
6. [6] TradeIndia / Univision Softech Pvt. Ltd. Machine Vision Ring Light, listed ₹12,000 (optional price reference, not used as the dome-light line item in the BOM). Open source page
7. [7] OpenCV documentation. Image filtering, colour conversion, thresholding, contours and connected components API. Open source page

Source discipline: price facts in the BOM are attributed to vendor listing [1] and [2]. Other BOM amounts are budget allowances only; they are not represented as researched vendor prices. Engineering recommendations and calculated values follow from the scenario data and formulas stated in this report.